

Mahboob Alam

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Professional Summary

Entry-level Machine Learning Engineer with strong hands-on experience building, evaluating, and deploying ML systems across computer vision, NLP, and structured data domains. Proven ability to translate research ideas into reproducible, deployment-ready pipelines with explicit engineering tradeoffs. Experienced with containerization, CI/CD workflows, REST APIs, and production-oriented ML infrastructure.

Technical Skills

Programming: Python, SQL

Machine Learning: Supervised Learning, Segmentation, Model Evaluation, Feature Engineering

Deep Learning: CNNs, Attention Mechanisms, Transformers, Lightweight / Efficient Models

NLP: Text Summarization, Sentiment Analysis, Information Extraction

Frameworks & Tools: PyTorch, Scikit-learn, Hugging Face Transformers, Pandas, NumPy

ML Engineering: FastAPI, REST APIs, Docker, CI/CD, Modular Training & Inference Pipelines

Visualization: Plotly, Matplotlib, Seaborn

Professional Experience

Research Intern

Jun 2025 – Present

Defence Institute of Advanced Technology (DIAT-DRDO), Pune

- Designed a Physics-Informed Attention U-Net for underwater image segmentation using explicit domain priors.
- Engineered a physics-aware feature extraction branch based on the Jaffe-McGlamery optical model to improve robustness in degraded underwater conditions.
- Integrated physics-guided features with attention mechanisms to enhance object boundary accuracy and reduce false positives.
- Built reproducible training and inference pipelines in PyTorch with controlled experiments and ablation tracking.
- Benchmarked against strong baselines (U-Net, Attention U-Net, DeepLabV3+) under identical evaluation protocols.
- Achieved state-of-the-art performance:
 - mIoU: 97.38%
 - Dice Score: 98.18%
 - Pixel Accuracy: 99.54%
- Performed systematic error analysis to identify failure modes and guide architectural refinements.
- Codebase: github.com/MahboobAlam0/piaunet

Projects

CI/CD Pipeline for Drone Autopilot Software using SITL-based Testing

Docker, GitHub Actions, CI/CD, Linux

- Built a containerized DevOps pipeline for ArduPilot to enable reproducible builds and automated testing.
- Designed Docker-based build environments to standardize toolchains and eliminate local setup inconsistencies.
- Implemented CI workflows using GitHub Actions to automate validation and ensure build stability.
- Structured a clear source → build → artifact pipeline aligned with production-grade software practices.
- Chose containerized builds over native local builds to prioritize reproducibility and CI reliability at the cost of longer build times.
- Codebase: github.com/MahboobAlam0/ardupilot_devops

FinSight: Financial Sentiment & Narrative Intelligence System

Python, FastAPI, Transformers, Docker, Streamlit

- Built a Dockerized end-to-end NLP system for business and financial news analysis.
- Designed a data ingestion → preprocessing → transformer inference → API response pipeline.
- Implemented abstractive summarization using DistilBART, trading minimal quality loss for reduced inference latency.
- Integrated multilingual sentiment analysis using BERT-based models for comparative market insights.
- Exposed REST APIs via FastAPI for batch and on-demand inference.
- Codebase: github.com/MahboobAlam0/finsight-ai

HMT-ECGNet: A Lightweight Hierarchical Multi-Lead ECG Classification on PTB-XL

Python, PyTorch, FastAPI, Docker

- Designed a lightweight hierarchical deep learning model for multi-label ECG classification.
- Achieved Macro-AUROC ≈ 0.92 and Macro-F1 ≈ 0.73 on the PTB-XL dataset.
- Reduced model size to **338K parameters**, enabling CPU-friendly inference (25 $\times$ fewer parameters than ResNet baselines).
- Built a Dockerized FastAPI service exposing REST endpoints for real-time ECG prediction.
- Explicitly traded minimal accuracy loss for lower latency and deployment feasibility.
- Codebase: github.com/MahboobAlam0/hmt_ecg_healthmonitoringsystem

Business Metric-Driven Churn Intervention System

Python, Pandas, Scikit-learn

- Built a churn prediction pipeline on structured customer data (7,000+ records).
- Integrated churn probability with customer lifetime value and intervention cost to guide retention decisions.
- Designed a cost-sensitive prioritization framework to maximize ROI under budget constraints.
- Conducted sensitivity analysis to validate robustness across varying business assumptions.
- Codebase: github.com/MahboobAlam0/businesschurn

Publications

Alam, M. et al., “Physics-Informed Attention U-Net (PIAU-Net): An Enhanced U-Net Framework for Underwater Image Segmentation,” *Indian Journal of Technical Education*, Vol. 48, No. 2, Dec 2025.

Education

M.Tech in Data Science

2024 – 2026

Defence Institute of Advanced Technology (DIAT-DRDO), Pune
CGPA: 8.10

B.Tech in Computer Science and Engineering

2019 – 2023

RBSSIET, Rewari